Assignment No:- 01

Title:- Write a program non-recursive and recursive program to calculate Fibonacci numbers and analyze their time and space complexity.

Program Code:-

1]Fibonacci[non-recursive]

```
COUNT = 0
x=int(input("Enter Number of Terms :"))
first=0
sec=1
c=0
if(x<0):
   print("Enter valid input..")
elif(x==0):
   print(0)
elif(x==1):
   print("Fibbonacci series upto",x,"is",first)
else:
   while c<x:
      print(first)
      COUNT = COUNT +1
      nth=first+sec
      COUNT = COUNT +1
      first=sec
      COUNT = COUNT +1
      sec=nth
      COUNT = COUNT +1
      c+=1
      COUNT = COUNT +1
print("Steps required using Counter ", COUNT)
```

OUTPUT:-

```
Enter Number of Terms :7
0
1
1
2
3
5
8
Steps required using Counter  35
```

2] Fibonacci[Recursive]

```
COUNT=0
def recur_fibo(n):
  global COUNT
  COUNT=COUNT+1
  if n <= 1:
     return n
  else:
     return(recur_fibo(n-1) + recur_fibo(n-2))
# take input from the user


nterms = int(input("How many terms? "))
# check if the number of terms is valid
if nterms <= 0:
  print("Plese enter a positive integer")
else:
  print("Fibonacci sequence:")
  for i in range(nterms):
     print(recur_fibo(i))


print("Steps reqired using Counter ", COUNT)
```

OUTPUT:-

```
How many terms:- 5
Fibonacci sequence:
0
1
1
2
3
Steps required using Counter  19
```